Responsive Design

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Responsive Design

- Definition
- Mobile 1st
- Viewport Size & Pixel Ratio
- Media Queries

Agenda

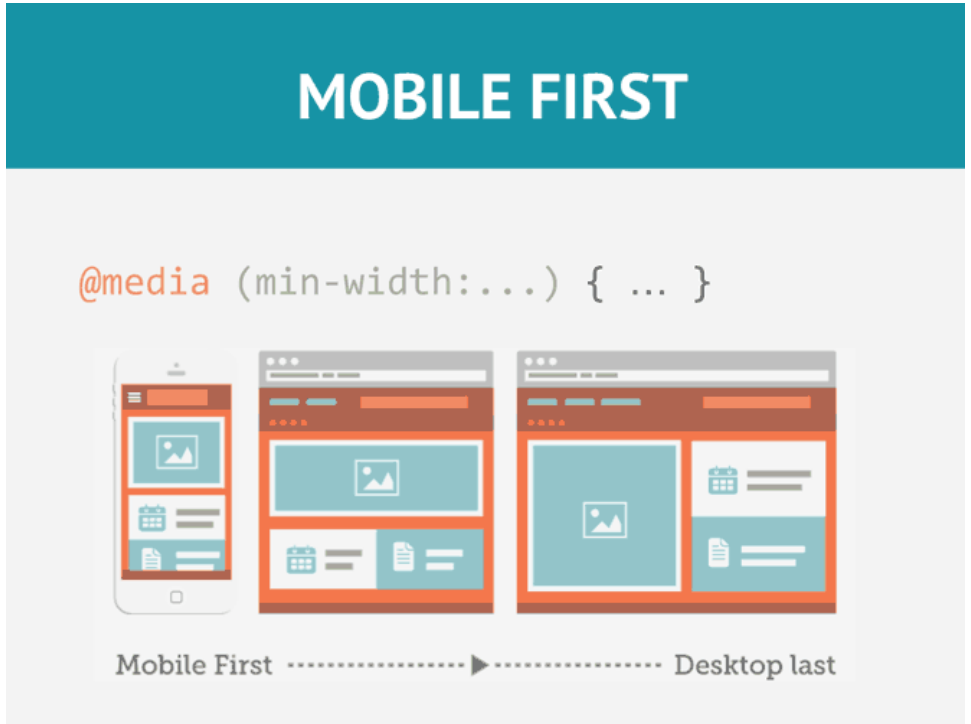
Responsive Design

– we want to ensure that

our Angular app looks good on any device

Mobile 1st

- Typically you start implementing the mobile version
  - more than 2/3 of access thru mobiles
- Use media queries for
  - tablet (e.g. 2 columns)
  - laptop (e.g. 3 columns)
  - desktop (e.g. 4 columns)



https://www.silocreativo.com/en/media-queries-css-work/

Viewport Size & Pixel Ratio

- Viewport Size
- Device Orientation
- Pixel Ratio
  - "Retina" === 2x → 4px for 1
  - nowadays 3x → 9px for 1
- https://www.mydevice.io/
  - https://yesviz.com/viewport/
  - <meta name="viewport" content="width=device-width, initial-scale=1" />

Media queries

– media queries are like if statements depending on viewport

MEDIATYPE

```
@media not|only mediatype
all
print
screen
speech
braile / projection / tv (Deprecated)
```

MEDIA FEATURE

```
@media and|not|only (media feature)
height
width
max-height / min-height
max-width / min-width
```

Media queries examples

- @media screen {}

- @media (min-width: 640px) {}

- @media (min-width: 640px) {}
- @media (min-width: 40em) {}

- @media (min-width: 40em) and (orientation: landscape) {}

https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_media_queries/Using_media_queries

Best Practices

- use relative units (em)
- mobile first, then use min-width
- make imags responsive (width 100%)
- test on different (native) devices

Tips

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Summary

Questions?

ANGULAR
ARCHITECTS